JAN-P-408

17 OCTOBER 1946

JOINT ARMY-NAVY SPECIFICATION

PENTOLITE, 50/50

This specification was approved by the War Department and the Navy Department for use of procurement services of the Army and the Navy and supersedes the following specification:

U. S. Army
50–11–126
24 Jan. 1944

A. APPLICABLE SPECIFICATIONS, OTHER PUBLICATIONS, AND DRAWINGS

A–1. *Specifications.*—The following specifications, of the issue in effect on date of invitation for bids, form a part of this specification:

JOINT ARMY-NAVY SPECIFICATIONS

JAN–P–387—Pentaerythrite Tetranitrate (PETN).
JAN–T–248—Trinitrotoluene (TNT).

U. S. ARMY SPECIFICATIONS

50–0–1—General Specification for Ammunition except Small Arms Ammunition.[1]
50–11–32—Acetone (Dimethyl-Ketone) (for Use in the Manufacture of Smokeless Powder).
100–2—Standard Specification for Marking Shipments by Contractors.[1]

NAVY DEPARTMENT SPECIFICATION

General Specifications for Inspection of Material.[2]

FEDERAL SPECIFICATION

RR–S–366—Sieves; Standard, Testing.

A–2. *Other publications.*—The following publications, of the issue in effect on date of invitation for bids, form a part of this specification:

ARMY ORDNANCE PUBLICATION

Packaging Instruction No. PIF–26—Shipping Container for Trinitrotoluene, etc.[1]

BUREAU OF SUPPLIES AND ACCOUNTS PUBLICATION

Navy Shipment Marking Handbook.[2]

INTERSTATE COMMERCE COMMISSION PUBLICATION

Regulations for Transportation of Explosives and Other Dangerous Articles, etc.

NATIONAL BUREAU OF STANDARDS PUBLICATION

Circular C434—Testing of Glass Volumetric Apparatus.

A–3. *Drawing.*—The following drawing, of the issue in effect on date of invitation for bids, forms a part of this specification:

[1] Applicable only to Army purchases.
[2] Applicable only to Navy purchases.

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ARMY ORDNANCE DRAWING

81–3–169—120° C. Vacuum Stability Test Bath.[3]

B. GRADE

B–1. This specification covers the following grades of pentolite 50/50, as specified in the contract or order (see par. H–2):

Grade I
Grade II

C. MATERIAL AND WORKMANSHIP

C–1. The raw materials used in the manufacture of pentolite 50/50 shall conform to the applicable specifications listed in paragraph A–1 and shall be of the grade or class specified as follows:

PETN—Class C, Joint Army-Navy Specification JAN-P–387.
TNT—Grade I, Joint Army-Navy Specification JAN–T–248.

C–2. Pentolite 50/50 shall consist of an intimate mixture of PETN and TNT in the proportions specified in paragraph D–2. It shall be made either by the co-precipitation method (dissolve the PETN and TNT in acetone, mix, and filter the solution, precipitate with clean water), or by the water slurry method (add a filtered acetone solution of TNT to a water slurry of stabilized and crystallized PETN under agitation).

D. GENERAL REQUIREMENTS

D–1. *Volatile matter.*—Maximum 0.5 percent.

D–2. *Composition.*—

	Percent
PETN	50 ± 2
TNT	50 ± 2

D–3. *Total acetone insoluble.*—Maximum 0.04 percent.

D–4. *Inorganic acetone insoluble.*—Maximum 0.02 percent.

D–5. *Acidity or alkalinity.*—Maximum 0.005 percent.

D–6. *Grit.*—None.

D–7. *PETN granulation.*[4]—After separation from TNT as specified in paragraph F–4g, the PETN shall conform to the following granulation requirements using U. S. Standard sieves conforming to the requirements of Federal Specification RR–S–366:

Through sieve No.	*Percent*
30	95 minimum
200	30 maximum

D–8. *Stability.*—When tested in accordance with paragraph F–4h not more than 5.0 ml of gas shall be liberated.

E. DETAIL REQUIREMENTS

E–1. *Bulk density.*—The bulk density of grade II pentolite 50/50 in gm. per ml. shall be a minimum of 0.60 and a maximum of 0.80.

F. METHODS OF SAMPLING, INSPECTION, AND TESTS

F–1. *Size of lot.*—Unless otherwise specified in the contract or order, a lot shall consist of not more than 50,000 pounds of pentolite 50/50.

[3] Applicable only to Army purchases.
[4] This requirement is not applicable to pentolite 50/50 manufactured by the water slurry method.

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F–2. *Sampling.*—

F–2a. *Primary sample basis.*—The Government inspector shall designate the number of containers, the location of the individual containers, and the method of sampling which shall be representative of the lot. In any case, not less than 10 containers shall be selected to represent the lot. The supplier shall be given no prior information to indicate which particular containers will be selected. A representative 1-pound primary sample shall be taken from each selected container. Each primary sample shall be mixed thoroughly, divided into two 8-ounce portions, and transferred to approved sample containers so labeled that the shipping containers from which these primary samples were taken can be identified readily. One 8-ounce portion shall be taken for laboratory examination and the other 8-ounce portion shall be retained for possible check analysis. Spot check samples shall be taken from time to time, from any portion of the lot, by the Government inspector.

F–2b. *Composite sample basis.*—When acceptance on basis of primary sample analysis is not feasible or advisable in the opinion of the Government inspector, the acceptance tests shall be made on a composite sample prepared by mixing a blend of equal portions taken from each of the primary samples and reducing the blend to obtain a composite sample weighing approximately 1 pound. The remaining portions of the primary samples shall be held by the Government inspector for possible future examination should the composite sample fail to meet the requirements of this specification.

F–2c. *Marking of sample containers.*—Each sample container shall be labeled to show the name of the material, manufacturer, plant, contract or order number, number of pounds in the lot, and lot number.

F–3. *Inspection.*—

F–3a. *Army.*—Inspection shall be made as specified in U. S. Army Specification 50–0–1.

F–3b. *Navy.*—Pentolite 50/50 shall be inspected at the place of manufacture unless otherwise specified in the contract or order.

F–3c. *Lot acceptance.*—Acceptance of the lot shall be based on the analysis of the individual primary samples or the composite samples selected to represent the lot. No lot shall be accepted when the analysis of any primary sample or the composite sample shows noncompliance in any respect with any requirement of this specification. Spot check samples taken from any portion of the lot shall comply with the requirements of this specification. Reports of acceptance shall show the maximum, the minimum, and the average of test results on all primary samples and spot check samples selected to represent the lot or block of lots. Each batch shall comply with the requirements of sections D and E. The contractor's record of batch analyses shall be available at all times for examination by the Government inspector.

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F–4. *Tests.*—The laboratory tests shall be made in accordance with the following paragraphs. For Navy purchases, the tests shall be made at a Government laboratory unless otherwise specified in the contract or order.

F–4a. *Volatile matter.*—Dry an accurately weighed sample of approximately 5 gm. for ½ hour in a desiccator containing concentrated H_2SO_4 and evacuated to an absolute pressure of not more than 360 mm. nor less than 100 mm. of mercury. The desiccator shall be equipped to prevent the sample from coming into contact with the H_2SO_4. Calculate the loss in weight of the sample as percentage of volatile matter.

F–4b. *Composition.*—

F–4b(1). *Saturated PETN solution.*—Prepare a saturated solution of PETN in chloroform at least 24 hours before use. The saturated solution shall be in contact with excess PETN. The portion of the saturated solution required for use shall be filtered just before the determination is made to assure saturation at the temperature at which the analysis is conducted.

F–4b(2). *Procedure.*—Transfer an accurately weighed portion of approximately 6 gm. of the sample to a tared, 30 ml. sintered glass crucible of medium porosity. Place the crucible on a filter adapter without suction. Add 10 ml. of the saturated PETN in chloroform solution. Allow to stand for 1 minute, then apply suction until the filtrate ceases to drip, and release the suction. Repeat to a total of five extractions. On the final extraction, hold the suction for a period of greater than 1 minute in order to aspirate. Dry to constant weight in a 100° C. oven and calculate the gain in weight as percentage of PETN. The percentage of TNT is found by subtracting from 100, the sum of the percentages of PETN and volatile matter.

F–4c. *Acetone insoluble (total).*—Dissolve a weighed portion of approximately 10 gm. of the sample in 50 ml. of acetone, boiling gently on a steam bath until solution is complete. Decant through a tared filtering crucible, transferring all insoluble matter to the crucible. Wash the latter with hot acetone, dry at 100° C. for 2 hours, cool in a desiccator and weigh. Calculate the gain in weight as percentage of acetone insoluble material.

F–4d. *Acetone insoluble (inorganic).*—This test need not be made if the acetone insoluble (total), paragraph F–4c, is less than 0.02 percent. Heat the crucible with insoluble residue (par. F–4c) over a Bunsen burner to remove all organic matter, cool in a desiccator and weigh. Calculate the weight of the residue in the crucible as percentage of inorganic matter.

F–4e. *Acidity or alkalinity.*—Stir a weighed portion of approximately 10 gm. of the sample vigorously with 50 ml. of boiling distilled water, until, as the mixture cools, the TNT granulates. When cool and the PETN has settled, decant off the supernatant liquid and extract the material with an additional 50 ml. of boiling distilled water. Repeat the stirring, cooling, settling and decantation. Titrate the combined aqueous extracts with N/50 sodium hydroxide or sulfuric acid using methyl red as the indicator. Correct the result by means of a blank determination. Calculate

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the acidity or alkalinity as percentage of sulfuric acid or sodium carbonate in the sample.

F–4f. *Grit.*—Place approximately 50 gm. of the sample in a beaker, add 50 ml. of acetone, cover the beaker and heat on a steam bath for about 10 minutes. Pour on a U. S. Standard No. 120 sieve, transfer all remaining insoluble material to the sieve and wash with hot acetone. Transfer the material retained on the sieve to a smooth glass slide. Rub the material on the glass slide by exerting pressure with a smooth steel spatula blade. Note if particles of grit are present as indicated by the persistence of a scratching noise when pressing and rubbing of the material on the glass is continued.

F–4g. *PETN granulation.*—Place a weighed portion of approximately 15 gm. of the sample in a 400 ml. beaker and add 70 to 80 ml. of chloroform saturated with PETN. (See par. F–4b(1).) Allow to stand at room temperature (20° to 25° C.), with occasional swirling, for about 10 minutes to dissolve the TNT. Filter through paper and remove as much chloroform from the PETN as practicable, by suction. Transfer the filter paper and its contents to a watch glass and place in a 100° C. oven for ½ hour. Tare weigh each of the appropriate 3-inch sieves specified in paragraph D–7 and nest these in the proper order in a Buchner funnel equipped with a filter paper. After cooling and weighing the PETN sample accurately, transfer it quantitatively to the top sieve. Gently wash the material back and forth across the sieve with a spray of 75 percent alcohol-water solution saturated with PETN until no more material goes through the sieve. Remove the top sieve and repeat the operation on each successive sieve using about a liter of alcohol-water solution for each sieve. Place each sieve and its contents on an asbestos paper tray and dry in a 100° C. oven for ½ hour. Before removing the material from the sieves, hold the sieve over a piece of paper and tap gently 10 times to sift through any fine adhering particles and add this to the next sieve. Weigh each sieve and contents and calculate the increase in weight to percentages of material retained by the different sieves and from these values calculate the percentage of the sample passing through each sieve. The amount passing through the 200-mesh sieve is calculated by difference.

F–4h. *100° C. vacuum stability test.*—

F–4h(1). *Test procedure.*—Transfer a 5-gm. sample, dried at 65° C. for 2 hours, to the heating tube of the apparatus shown on figure 1. Connect the capillary tube to the heating tube and seal the connection with 1 ml. of mercury. Clamp the apparatus so that the long section of the capillary tube is in a nearly vertical position and the lower end rests on a solid support. Transfer 7.0 ml. of mercury to the cup at the lower end of the capillary tube. Connect a vacuum pump to the lower end of the capillary tube and evacuate the system until the pressure is reduced to approximately 5 mm. of mercury. Disconnect the pump and measure the total vertical height of the column of mercury in the capillary tube. Measure and subtract the vertical height of the mercury in the cup. The difference shall be represented by the symbol H_1. Note

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the room temperature (t_1) and the barometric pressure. Subtract the value H_1 from the barometric pressure in mm. Represent this difference by the symbol P_1. Insert the heating tube in a constant temperature bath (Army installations use apparatus described on Army Ordnance Drawing 81–3–169) containing a mixture of approximately 2 parts of toluene and 1 part of xylene. Maintain at a temperature of 100.0° ± 0.5° C. for 40 hours. Remove the heating tube from the bath and allow it to cool to room temperature. Measure the total vertical height of the column of mercury in the capillary tube and subtract the vertical height of the mercury in the cup. This difference shall be represented by the symbol H. Note the room temperature (t) and the barometric pressure. Subtract the value H from the barometric pressure in mm. Represent this difference by the symbol P.

F–4h(2). *Calibration of glass tube.*—Determine the volume in ml. of the 15.5 cm. heating tube by running in mercury from a buret until the tube is filled to the level at which the ground glass joint of the capillary tube will make contact with the mercury. Subtract from the indicated buret reading, the volume of explosive used in the test (5 ml.). The difference shall be represented by the symbol A. Transfer 7.0 ml. of mercury to the cup at the lower end of the capillary tube. Clamp the tube in an upright vertical position, and measure the height in mm. of the mercury column in the capillary tube (approximately 25 mm.). Measure the length in mm. of each of the 3 parts of the capillary tube and add these values to obtain total length. From the total length subtract the height of the mercury column in the capillary tube as previously obtained. Represent this difference by the symbol B_1. From the total length subtract the height of the column of mercury in the capillary tube measured at the end of the test described in paragraph F–4h(1). Represent this difference by the symbol B. Determine the capacity of the capillary tube per unit of length as follows: Transfer an accurately weighed sample of approximately 10 gm. of mercury to the cup at the lower end of the capillary tube. Manipulate the tube so that when it is horizontal, mercury is contained in a continuous section of the longest part of the tube and measure the length of the mercury column. Repeat this twice with the mercury in 2 other parts of the long section of the tube. Calculate the average of the 3 measured lengths of the mercury column. Represent the unit capacity in ml. per mm. of the capillary tubing by the symbol C. This can be obtained from the formula.

$$C = \frac{W}{DL}$$

where

C = unit capacity of capillary tubing in ml. per mm.
W = grams of mercury
D = density of mercury at temperature of determination
L = average measured lengths of mercury column in mm.

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F–4h(3). *Calculation of liberated gas volume.*—Calculate the volume of gas in ml., at standard conditions, liberated in the test described in paragraph F–4h(1) using the values represented by the symbols described in the preceding paragraphs in the following formula:

Volume of gas =

$$[A + C\,(B - H)]\left\{\frac{273\,P}{760\,(273 + t)}\right\} - [A + C\,(B_1 - H_1)]\left\{\frac{273\,P_1}{760\,(273 + t_1)}\right\}$$

F–4i. *Bulk density.*—Pour 50 gm. of the sample into a 100 ml. graduated glass measuring cylinder, replacing the glass-stopper with a cork. Drop the cylinder vertically from a height of 2½ inches onto a piece of hard leather 50 times. Level the surface by gentle side tapping and read the volume in ml. Calculate the bulk density in gm. per ml. The 100 ml. cylinder must be precision grade, meeting the requirements for accuracy as specified in National Bureau of Standards Circular C–434 (tolerance in 100 ml. of ± 0.20 ml.), and marked with 1 ml. subdivisions. The length between the 10 ml. and 100 ml. marks must be 145 ± 15 mm.

F–5. *Resubmission.*—If the lot or any portion of the lot fails to pass the inspection tests, the manufacturer shall have the option of having additional analyses made without expense to the Government to affect complete segregation of all defective portions of the lot. The manufacturer may then remove or replace defective portions of the lot and resubmit the lot for acceptance provided that the markings on the containers are such that complete removal or replacement of defective portions of the lot can be made to the satisfaction of the Government inspector.

G. PACKAGING, PACKING, AND MARKING FOR SHIPMENT

G–1. *Packaging and packing.*—

G–1a. *General.*—The material shall be packed in rectangular boxes or fiber drums of 50 pounds capacity, lined with strong paper bags, and so constructed as to comply in all respects with the requirements of Interstate Commerce Commission regulations.

G–1b. *Army.*—In addition to the requirements of paragraph G–1a, packaging for the Army shall comply with the requirements of Army Ordnance Packaging Instructions No. PIF–26.

G–2. *Marking.*—In addition to any special marking required by the contract or order, shipments for the Army shall be marked in accordance with the requirements of U. S. Army Specification 100–2; for the Navy in accordance with the requirements of the Navy Shipment Marking Handbook.

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H. NOTES

H–1. *Use.*—Pentolite 50/50, grade I, is intended for use in cast loaded ammunition. Pentolite 50/50, grade II, is intended for use in pellets and press loaded ammunition.

H–2. Requests, requisitions, schedules, and contracts or orders should state the title of the specification, the number and date, and the grade required. (See par. B–1.)

H–3. Copies of Army Ordnance Packaging Instructions No. PIF–26 and Ordnance Department Drawing 81–3–169 may be obtained from the Industrial Service, Ammunition Division, Office, Chief of Ordnance, War Department, Washington 25, D. C.

H–4. Copies of National Bureau of Standards Circular C434 may be obtained upon application, accompanied by postal note, money order, coupon or cash, to the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Price: 10 cents.

H–5. Copies of Interstate Commerce Commission Regulations for Transportation of Explosives and other Dangerous Articles, etc., may be obtained upon application, accompanied by postal note, money order, coupon, or cash, to the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Price: 40 cents.

H–6. Copies of Joint Army-Navy specifications and Federal specifications (required for Army purchases) and U. S. Army specifications may be obtained, as indicated in the "Index of United States Army, Joint Army-Navy, and Federal Specifications Used by the War Department." Copies of this Index may be obtained from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Agencies within the War Department will obtain copies of Joint Army-Navy, United States Army, and Federal specifications through established War Department channels. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

H–7. Copies of Joint Army-Navy specifications and Federal specifications (required for Navy purchases), Navy Department specifications and the Navy Shipment Marking Handbook may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that Naval activities should make application to the Supply Officer in Command, Naval Supply Depot, Bayonne, N. J. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

H–8. Copies of this Joint Army-Navy specification (required for Army purchases) may be obtained from the Office, Chief of Ordnance, War Department, Washington 25, D. C.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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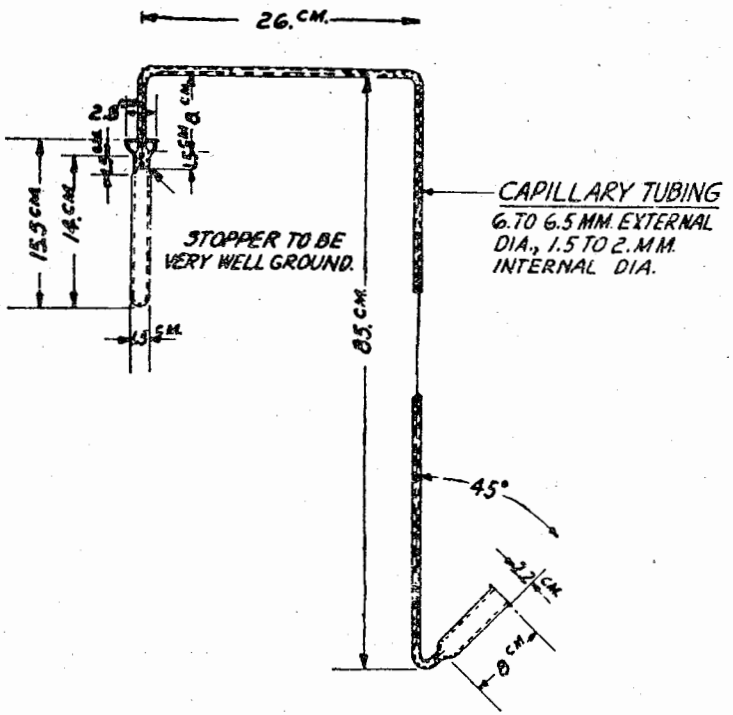


FIGURE 1.—*Apparatus for 100° C. vacuum stability test.*

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